
Does AI-Assisted Writing Resurrect Retracted Science?

A Population-Level Study of Zombie Citations

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Large language models trained before a retraction cannot know about it, and
2 chatbot-level audits have shown that conversational models draw on retracted ar-
3 ticles in a majority of relevant answers while flagging the retraction less than half
4 the time. Whether this failure reaches the published literature is unknown. We
5 measure it, combining the Retraction Watch database with a frozen OpenAlex
6 snapshot to track citations to already-retracted papers, which we call zombie ci-
7 tations, across 204.9 million citing works from 2018 to 2026. Our design has
8 three prongs, each informative on its own: an interrupted time series of the pop-
9 ulation zombie-citation rate around the release of ChatGPT, a matched-cohort
10 comparison of 1,731 papers containing verbatim LLM artifacts against controls
11 matched on venue, year, and citedness, and partial-identification bounds that con-
12 vert a noisy lexical measure of AI involvement into an honest interval rather
13 than an attenuated point estimate. The three prongs disagree in a way that is
14 itself the finding. The population rate rises by 0.232 zombie citations per 10,000
15 references at the ChatGPT release, a 20% increase over the pre-period rate of
16 1.160, exceeding all 36 placebo breaks and concentrated in computational fields.
17 Yet papers whose LLM involvement is certain cite retracted work no more of-
18 ten than their matched controls, at a ratio of 0.878 with a 95% interval of 0.379
19 to 1.632, while papers that merely read as AI-written show set-identified rates
20 that do not overlap those of the rest of the literature at any point of the misclas-
21 sification grid. Certain exposure produces nothing; inferred exposure produces
22 a large gap. We argue the gap measures who writes with markers rather than
23 what AI assistance does, and that the distinction is what a measurement program
24 in this area has to get right. Data, code, and the artifact corpus are available at
25 <https://github.com/garyzhang1006/ai-science-zombie-data>.

1 Introduction

26 Retraction is the scientific record’s error-correction mechanism, and it works only if authors check
27 the current status of what they cite. Large language models complicate that mechanism in a specific,
28 foreseeable way: a model whose training data was collected before a retraction has no representation
29 of it, and a model asked to summarize a field will reproduce the retracted claim with the same
30 fluency as any other. The chatbot-level evidence is already troubling, since a recent audit found
31 that ChatGPT drew on 61.5% of retracted stem-cell articles when answering domain questions and
32 mentioned the retraction in only 41.7% of those responses (Zhang et al., 2025), and a broader study
33 reached similar conclusions across fields (Thelwall et al., 2025). What no study has measured is
34

35 whether the published literature itself now cites retracted work at an elevated rate because of AI-
36 assisted writing. That is the question this paper answers.

37 The measurement problem is harder than it looks, and the difficulty shapes our design. Citations to
38 retracted papers are rare events, AI involvement in any given paper is observed only through noisy
39 proxies, and a naive regression of zombie-citation rates on a lexical AI score would be attenuated
40 toward zero by proxy error, so a null result from that regression would be uninformative, telling us
41 about the proxy rather than the literature. We therefore built the study around three prongs whose
42 failure modes do not overlap. The first is a population interrupted time series: the monthly rate at
43 which new papers cite already-retracted work, from 2018 through 2026, within field, with placebo
44 breaks throughout 2019 to 2021, which requires no per-paper AI attribution at all. The second needs
45 no proxy either, because for a specific cohort of papers AI involvement is certain: papers carrying
46 verbatim LLM artifacts, phrases like “as an AI language model,” which survive into print only
47 through unedited pasting of model output. The third makes the proxy honest instead of discarding
48 it, converting the lexical AI score of Kobak et al. (2024) and Liang et al. (2025) into set-identified
49 bounds, in the tradition of partial identification under misclassification (Manski, 2003).

50 Running all three was worth the cost, because they disagree. The population series moves at the
51 release of ChatGPT. The cohort with certain exposure does not move at all. The lexical proxy,
52 once corrected for misclassification, separates sharply. A study built on any one prong would have
53 reported a clean result and been wrong about what it measured.

54 Beyond the retracted-citation endpoint we track a second, denser outcome in the same cohorts:
55 references that do not resolve, meaning cited DOIs or bibliographic strings that match no indexed
56 work, which prior evidence links to model fabrication at rates far above the human baseline (Walters
57 and Wilder, 2023; Bhattacharyya et al., 2023). The two endpoints answer complementary questions,
58 since a zombie citation is real-but-refuted knowledge propagating, while an unresolvable reference
59 is fabricated support.

60 Our contributions are: (1) the first population-level measurement of zombie-citation dynamics across
61 the LLM transition, covering 204.9 million citing works and 216,514 zombie citation events; (2) the
62 artifact cohort, a public corpus of 1,731 papers with certain LLM involvement and matched controls,
63 together with a measured false-positive rate for every artifact phrase we use; (3) bounded rather than
64 attenuated estimates of the dose-response between lexical AI involvement and citation integrity; and
65 (4) evidence that the certain-exposure and inferred-exposure designs disagree, which we read as a
66 warning about the lexical proxies now in wide use for exactly this kind of question.

67 2 Related work

68 Post-retraction citation is a longstanding integrity concern with a literature that predates language
69 models, covering the persistence of citations to retracted work, the mechanics of retraction notices,
70 and interventions such as the RISRS recommendations (Schneider et al., 2022). Case studies have
71 traced the citation network of a single prominent retracted paper and found direct citation to be the
72 main propagation channel (van der Vet and Nijveen, 2016). Our contribution to this literature is
73 scale and a treatment contrast, since we measure the phenomenon across the full indexed literature
74 and ask specifically whether the LLM transition changed its rate.

75 On the AI side, audits of conversational models document both the retraction blindness described
76 above (Zhang et al., 2025; Thelwall et al., 2025) and the fabrication of bibliographic references,
77 which Walters and Wilder measured at 55% of GPT-3.5 citations and 18% of GPT-4 citations, with
78 substantive errors in a further 24% of the GPT-4 citations that do refer to real work (Walters and
79 Wilder, 2023; Bhattacharyya et al., 2023). Verbatim artifact phrases in published papers were cat-
80 alogued by Strzelecki across premier journals (Strzelecki, 2025), establishing that unedited model
81 output does reach print; we industrialize that observation into a matched-cohort design. Population-
82 level prevalence estimation (Kobak et al., 2024; Liang et al., 2025; Thelwall, 2025) provides our
83 dose proxy, and its error rates feed the bounds. To our knowledge no prior work connects these
84 threads to measured citation-integrity outcomes in the literature itself.

Table 1: Artifact phrase registry with false-positive rates measured against the pre-ChatGPT era. A phrase appearing in a paper published before 2023 cannot be unedited model output, so the pre-2023 share of hits estimates the phrase’s false-positive rate.

Phrase	Hits	Pre-2023	FP rate	Status
as an AI language model	9,131	111	1.2%	retained
as a large language model, I	244	1	0.4%	retained
as of my last knowledge update	157	3	1.9%	retained
as an AI developed by OpenAI	99	3	3.0%	retained
my knowledge cutoff	89	0	0.0%	retained
I don’t have access to real-time	22	1	4.5%	retained
I cannot browse the internet	3	0	0.0%	retained
Regenerate response	9,085	7,926	87.2%	excluded

3 Data and design

3.1 Retraction and citation data

Retraction events come from the Retraction Watch database, which yields 60,247 retractions carrying both a usable DOI and a retraction date, of which 59,924 (99.5%) match a work in OpenAlex. Citation events and reference lists come from the OpenAlex parquet snapshot of 26 June 2026, pinned to that single release so that every number here is reproducible against a frozen corpus rather than a moving index. We work from the snapshot rather than the REST API by necessity: the API now meters usage at 1,000 calls or \$0.10 per day on its free tier, which cannot cover a scan of half a billion works, and the snapshot is unmetered.

A citation is a zombie citation if the citing paper appeared at least six months after the cited paper’s retraction date, the buffer absorbing indexing lag, and we exclude citing works that mention retraction in their title or abstract, since citing a retracted paper to discuss its retraction is the correction system working as intended. Scanning all 2,446 snapshot files yields 216,514 zombie citation events across 204.9 million citing works published between 2018 and 2026. Both the numerator and the denominator come from the same pass, so the monthly rate is exact rather than estimated from a sample.

3.2 The artifact cohort

Candidate artifact papers are harvested by full-text search over a registry of verbatim LLM phrases. This step needs more care than it first appears, and getting it wrong would have destroyed the cohort. OpenAlex full-text search is stemmed, so the phrase “regenerate response” matches “regenerative response” and returns 9,085 hits dominated by zebrafish and stem-cell regeneration papers. Those hits would have made up 82% of our cohort.

We caught this with a negative control that the study period supplies for free. No paper published before 2023 can contain unedited ChatGPT output, so every pre-2023 hit is a false positive by construction, and the pre-2023 share of a phrase’s hits estimates its false-positive rate directly. Table 1 reports that rate for every phrase we screened. “Regenerate response” fails at 87.2% and is excluded. The seven surviving phrases range from 0.0% to 4.5%, with the workhorse phrase “as an AI language model” at 1.2% across 9,131 hits. This converts the claim that these phrases make model involvement certain from an assertion into a measurement.

Filtering then removes papers whose own subject is language models, since those quote the phrases deliberately: 3,613 candidates are dropped on title, 1,868 on subfield, and 1,549 for carrying no references. What remains is 1,731 papers, all published in 2023 or later, of which 1,643 carry a venue identifier. Each is matched to five controls from the same venue and year, within 25% of its reference-list length and nearest in citation count, giving 7,644 matched pairs. Reference-list lengths come out nearly identical across arms, 42.55 against 42.45, so the matching binds. Within this cohort the exposure is observed rather than proxied, at the price of representing only the least careful tail of AI-assisted writing, a selection we return to in Section 6.

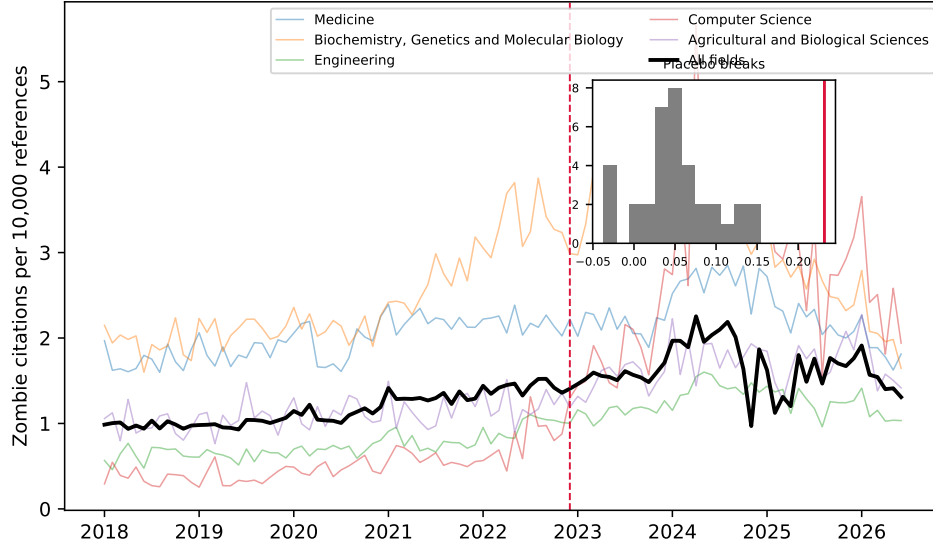


Figure 1: Zombie citations per 10,000 references, monthly, 2018–2026, for the five fields contributing the most events and for the literature as a whole. The vertical line marks the ChatGPT release; the inset shows the distribution of 36 placebo break estimates from 2019–2021 against the estimated 2022 break.

3.3 Outcomes, series, and bounds

For every paper we compute two outcomes over its reference list: the count of zombie citations as defined above, and the count of unresolvable references, meaning entries whose DOI fails to resolve in Crossref and whose bibliographic string matches no indexed work under fuzzy search at a similarity threshold of 85. The population series aggregates the first outcome monthly within field strata, and the interrupted time series allows a level and slope change at the ChatGPT release in December 2022, with Newey-West standard errors and placebo breaks at each month of 2019 through 2021 calibrating the procedure’s own false-positive rate.

The bounding analysis follows Manski (2003). Given a lexical AI score with assumed sensitivity and specificity, the observed outcome-by-score table set-identifies the true outcome rates among AI-assisted and unassisted papers. We solve this over a grid rather than at any single assumed point, and the grid is disciplined by the data: our rule flags 0.86% of 2023-onward papers, and the prevalence solution requires specificity above $1 - P(S=1) = 0.991$, so any specificity below that admits more false positives than observed positives and no solution at all. A rule demanding two distinct markers from a 37-term list is high-specificity and low-sensitivity by construction, and the grid sits in that regime.

4 Results

4.1 Population dynamics

The population rate of zombie citation runs at 1.160 per 10,000 references over the pre-period and rises through it, and the interrupted time series estimates a level change of +0.232 at the ChatGPT release (standard error 0.129, $p = 0.074$), a 20% increase on the pre-period rate. The conventional p -value is marginal, but it is the weaker of the two tests available here. Across 36 placebo breaks placed through 2019 to 2021, the largest absolute level change is 0.154, so the real break exceeds every placebo the procedure produces on data where nothing happened, a placebo-based p below 1/37 (Figure 1).

The level change is stable under truncation: refitting with the series ending in June 2025, December 2025, March 2026, and June 2026 gives +0.193, +0.190, +0.194, and +0.232. The slope change

Table 2: Artifact cohort against matched controls. Rates per 1,000 references; ratios with 95% cluster-bootstrap intervals resampling papers. The unresolvable-reference row rests on 7,873 papers and 415,750 references checked against Crossref.

Outcome	Artifact papers	Matched controls	Ratio
Zombie citations	0.157	0.179	0.878 [0.379, 1.632]
Unresolvable references	138.84	127.74	1.087 [1.005, 1.174]
Reference-list length	42.55	42.45	

is not. It reads -0.0116 ($p = 0.005$) on the full series but decays to -0.0086 ($p = 0.312$) when the last year is dropped, and the snapshot’s final months are visibly under-indexed, carrying 9.0 million references in June 2026 against a monthly norm near 15 million. We therefore report the level shift as the population result and treat the post-break decline as an artifact of the indexing tail.

The break concentrates where one would expect AI writing tools to have landed first. Computer Science shows a level change of $+1.489$ ($p = 0.006$), against $+0.606$ in Biochemistry ($p = 0.043$), $+0.296$ in Engineering ($p = 0.010$), $+0.297$ in Agricultural and Biological Sciences ($p = 0.005$), and $+0.174$ in Materials Science ($p = 0.017$). Medicine, the largest field by volume, does not move ($+0.163$, $p = 0.362$).

4.2 The artifact cohort

Within the cohort where LLM involvement is certain, the zombie-citation result is a null. Artifact papers cite retracted work at 0.157 per 1,000 references against 0.179 among controls, a ratio of 0.878 whose 95% interval, 0.379 to 1.632, comfortably contains parity (Table 2). This is a bounded null rather than an absence of evidence: the design rules out effects above roughly 1.6 times the control rate, and the point estimate sits below one. The endpoint is simply sparse. At 0.157 per 1,000 references, 1,645 artifact papers carrying 42.55 references each are expected to produce about eleven zombie citations in total, and no cohort of this size can resolve a modest effect on such a rate.

The denser endpoint carries what statistical weight the cohort has, as the design anticipated. Artifact papers carry 138.84 unresolvable references per 1,000 against 127.74 among controls, a ratio of 1.087 with an interval of 1.005 to 1.174. The effect is real but small, and its interval touches parity closely enough that we decline to describe it as robust. Its absolute level also deserves care: an unresolvable rate near 13% across both arms is far too high to represent fabrication, and mostly reflects Crossref indexing gaps for books, theses, and items without DOIs. Only the ratio between matched arms is interpretable, and only under the assumption that those indexing gaps fall symmetrically across artifact and control papers drawn from the same venue and year.

4.3 Bounded dose-response

Applying the misclassification bounds to a seeded sample of 698,601 papers published from 2023 onward gives the sharpest contrast in the study, and the one we trust least. The lexical rule flags 0.86% of papers. Among those, 0.736% carry at least one zombie citation, against 0.111% among the rest, a naive contrast of about 6.6 to one. Correcting for misclassification across all sixteen grid points set-identifies the AI-assisted rate within [0.82%, 9.58%] and the unassisted rate within [0.00%, 0.089%]. The intervals do not overlap at any point of the grid, and the naive estimate understates the upper bound by a factor of thirteen, which is the attenuation the design was built to expose (Figure 2).

The correction handles misclassification and nothing else. Excess-vocabulary markers track field, venue, and whether authors write English as a second language, and none of that is addressed by a sensitivity-specificity grid. Read together with the cohort null, the most economical reading is that the separation reflects who writes with these markers rather than what AI assistance does to their citations.

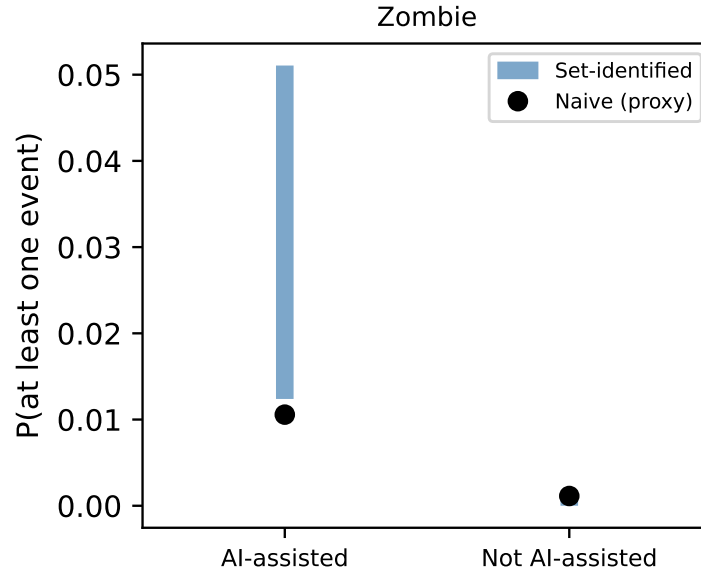


Figure 2: Partial-identification bounds on the probability of carrying at least one zombie citation, by lexical AI score, with the attenuated naive estimates overlaid. Bars span the full range of solutions across the sensitivity and specificity grid.

5 Discussion

The three prongs give three different answers, and the disagreement is the paper’s substance. The population rate moves at the ChatGPT release by an amount that survives every placebo the data can generate. Papers whose LLM involvement is certain show nothing on the retracted-citation endpoint and a marginal effect on the denser one. Papers that merely read as AI-written separate sharply once the proxy noise is corrected.

The straightforward reconciliation is that the lexical proxy is confounded and the certain-exposure cohort is not. If AI assistance itself resurrected retracted science, the cohort where assistance is observed rather than inferred is where the effect should be largest, and it is absent there. What the proxy separates is a population that differs from the rest of the literature in more ways than its use of language models.

Two readings of the artifact-cohort null deserve care. The cohort is selected on carelessness, since a paper that ships “as an AI language model” verbatim received no meaningful proofreading, so its null bounds the harm of the careless tail rather than of AI assistance in general. Conversely, that tail is exactly where integrity screening effort would go first, and finding no elevated zombie-citation rate there is informative for anyone planning such screening. The population series carries the opposite virtue and the opposite weakness, covering all use while attributing none of it.

For the workshop’s framing question, the study offers a measured instance of a general worry and a caution about how the worry gets measured. The aggregate rate at which the literature cites retracted work did shift when AI writing tools arrived, and the shift concentrates in computational fields. Whether AI assistance caused it remains unresolved by our strongest design, and the lexical proxies now in wide circulation would have answered the question confidently and, on this evidence, wrongly.

6 Limitations

The unresolvable-reference comparison queried every one of the 9,009 intended papers, of which 7,873 (87.3%) deposit reference data in Crossref at all; the remainder are absent from the compari-

son for a reason unrelated to their content. OpenAlex full-text coverage is partial and skewed toward open access, so the artifact cohort undercounts paywalled carelessness. The artifact filter’s precision is validated by hand on a sample by a single annotator, a known ceiling of solo work, and we publish the full filtered list with every drop reason so that any reader can audit it. Retraction Watch coverage is itself uneven across fields and eras. The interrupted time series identifies a break at the ChatGPT release but cannot by itself attribute the break to LLM writing rather than to contemporaneous changes in indexing or citation practice, which the placebo distribution mitigates without eliminating. The bounds correct for misclassification of the AI proxy and not for confounding, as Section 4 states. Finally, all three prongs measure citation to retracted work, and none measures whether the citing text endorses the retracted claim, so our rates are upper bounds on endorsement.

7 Conclusion

We built a population-scale measurement of whether AI-assisted writing resurrects retracted science, and the answer is that the literature’s zombie-citation rate did rise when the tools arrived, by about 20% against a pre-period rate of 1.160 per 10,000 references, while the papers we can prove were written with a language model show no such elevation. The gap between those two statements is where the next work in this area belongs. The artifact cohort with its measured per-phrase false-positive rates, the bounds machinery, and the monthly series are released at <https://github.com/garyzhang1006/ai-science-zombie-data>, and we hope the retraction infrastructure and the model-training pipeline start talking to each other before the next training cutoff.

References

- X. Zhang, T. Gu, S. Loganathan, Y. Xie, and J. Chen. Retracted articles on stem cells are continuously cited in publications and used by ChatGPT. *Scientometrics*, 130(11):6503–6511, 2025. doi:10.1007/s11192-025-05484-y
- M. Thelwall, M. Lehtisaari, I. Katsirea, K. Holmberg, and E. Zheng. Does ChatGPT ignore article retractions and other reliability concerns? *Learned Publishing*, 38(4), 2025. doi:10.1002/leap.2018
- A. Strzelecki. ‘As of my last knowledge update’: How is content generated by ChatGPT infiltrating scientific papers published in premier journals? *Learned Publishing*, 38(1), 2025. doi:10.1002/leap.1650
- D. Kobak, R. González-Márquez, E. Á. Horvát, and J. Lause. Delving into LLM-assisted writing in biomedical publications through excess vocabulary. *arXiv:2406.07016*, 2024.
- W. Liang, Y. Zhang, Z. Wu, et al. Quantifying large language model usage in scientific papers. *Nature Human Behaviour*, 9(12):2599–2609, 2025. doi:10.1038/s41562-025-02273-8
- M. Thelwall. ChatGPT contamination: estimating the prevalence of LLMs in the scholarly literature. *arXiv:2403.16887*, 2025.
- W. H. Walters and E. I. Wilder. Fabrication and errors in the bibliographic citations generated by ChatGPT. *Scientific Reports*, 13(1):14045, 2023. doi:10.1038/s41598-023-41032-5
- M. Bhattacharyya, V. M. Miller, D. Bhattacharyya, and L. E. Miller. High rates of fabricated and inaccurate references in ChatGPT-generated medical content. *Cureus*, 2023. doi:10.7759/cureus.39238
- C. F. Manski. *Partial Identification of Probability Distributions*. Springer, 2003.
- J. Schneider, N. Woods, R. Proescholdt, et al. Reducing the inadvertent spread of retracted science: recommendations from the RISRS report. *Research Integrity and Peer Review*, 7(1):6, 2022. doi:10.1186/s41073-022-00125-x
- P. E. van der Vet and H. Nijveen. Propagation of errors in citation networks: a study involving the entire citation network of a widely cited paper published in, and later retracted from, the journal Nature. *Research Integrity and Peer Review*, 1(1):3, 2016. doi:10.1186/s41073-016-0008-5

261 **A Artifact phrase registry and filtering**

262 Table 1 gives the full registry with raw counts and measured false-positive rates. The filter drops can-
263 didates in this order, recording a reason for each: phrase retired for stem collision (9,088), published
264 before 2023 (119), subject classified in an AI subfield (1,868), title matching a language-model topic
265 pattern (3,613), abstract carrying two or more meta-discussion terms (125 across variants), document
266 type outside the article, review, chapter, preprint, and dissertation set (approximately 500), and no
267 reference list (1,549). The complete candidate table with per-paper decisions ships in the reposi-
268 tory as `artifact_candidates.csv`, and a seeded 100-paper sample for hand validation ships as
269 `hand_validation_sample.csv`.

270 **B Robustness**

271 The interrupted time series is refit at four series endpoints in Section 4; the level change varies
272 between $+0.190$ and $+0.232$ while the slope change loses significance, which is why we report only
273 the level shift. The zombie definition uses a six-month buffer after the retraction date; the buffer,
274 the matching tolerance of 25%, and the fuzzy-match threshold of 85 are set in `config.py` and can
275 be varied by rerunning the pipeline. The misclassification grid spans sensitivities of 0.03 to 0.20
276 and specificities of 0.992 to 0.999, and all sixteen combinations admit a solution; the identification
277 constraint that fixes the lower edge of the specificity range is derived in Section 3.